

Exercise C2

Q1. Let p and q be the two roots of the equation, $mx^2 + x(2 - m) + 3 = 0$. Let m_1, m_2 be the two values of m satisfying $\frac{p}{q} + \frac{q}{p} = \frac{2}{3}$. Determine the numerical value of $\frac{m_1}{m_2^2} + \frac{m_2}{m_1^2}$.

Ans. 99

Sol. $mx^2 + x(2 - m) + 3 = 0 \begin{cases} p \\ q \end{cases}$

$$p + q = \frac{m - 2}{m}, pq = \frac{3}{m}$$

$$\Rightarrow p^2 + q^2 = (p + q)^2 - 2pq$$

$$\Rightarrow p^2 + q^2 = \frac{(m - 2)^2}{m^2} - \frac{6}{m}$$

$$= \frac{(m - 2)^2 - 6m}{m^2}$$

$$= \frac{m^2 + 4 - 10m}{m^2}$$

$$\Rightarrow \frac{p}{q} + \frac{q}{p} = \frac{2}{3}$$

$$\frac{p^2 + q^2}{pq} = \frac{2}{3}$$

$$3 \cdot (p^2 + q^2) = 2pq$$

$$3 \cdot \frac{(m^2 + 4 - 10m)}{m^2} = 2 \cdot \frac{3}{m}$$

$$m^2 + 4 - 10m = 2m$$

$$m^2 - 12m + 4 = 0 \begin{cases} m_1 \\ m_2 \end{cases}$$

$$m_1 + m_2 = 12, m_1 \cdot m_2 = 4$$

$$\Rightarrow m_1^3 + m_2^3 = (m_1 + m_2)^3 - 3m_1m_2(m_1 + m_2)$$

$$= (12)^3 - 3 \times 4 \times 12$$

$$= 12^3 - 12 \times 12$$

$$= 12^2(12 - 1)$$

$$= 144 \times 11$$

$$\Rightarrow \frac{m_1}{m_2^2} + \frac{m_2}{m_1^2}$$

$$\begin{aligned}
&= \frac{m_1^3 + m_2^3}{m_1^2 \cdot m_2^2} \\
&= \frac{144 \times 11}{4 \times 4} \\
&= 99
\end{aligned}$$

Q2. Solve for x the following equation:

$$\log_{(2x+3)}(6x^2 + 23x + 21) = 4 - \log_{(3x+7)}(4x^2 + 12x + 9)$$

Ans. -

Sol. -

Q3. Find the values of 'a' for which $-3 < [(x^2 + ax - 2)/(x^2 + x + 1)] < 2$ is valid for all real x.

Ans. $-2 < a < 1$

Sol. -

Q4. Find the minimum value of
$$\frac{\left(x + \frac{1}{x}\right)^6 - \left(x^6 + \frac{1}{x^6}\right) - 2}{\left(x + \frac{1}{x}\right)^3 + x^3 + \frac{1}{x^3}}$$
 for $x > 0$.

Ans. -

Sol. -

Q5. Find all numbers p for each of which the least value of the quadratic trinomial $4x^2 - 4px + p^2 - 2p + 2$ on the interval $0 \leq x \leq 2$ is equal to 3.

Ans. -

Sol. -

Q6. Find the values of 'b' for which the equation $2\log_{\frac{1}{25}}(bx + 28) = -\log_5(12 - 4x - x^2)$ has only one solution.

Ans. -

Sol. -

Q7. Solve the system of equations:

$$\log_a x \log_a (xyz) = 48, \log_a y \log_a (xyz) = 12, \log_a z \log_a (xyz) = 84 \text{ where } a > 0, \neq 1$$

Ans. -

Sol. -

Q8. Let $y = \sqrt{\log_2 3 \cdot \log_2 12 \cdot \log_2 48 \cdot \log_2 192 + 16} - \log_2 12 \cdot \log_2 48 + 10$. Find $y \in \mathbb{N}$.

Ans. $y = 6$

Sol. -

Q9. Find 'a' for which the equation $x^2 + (a - b)x + (1 - a - b) = 0$ has two distinct and unequal roots for $\forall b \in \mathbb{R}$?

Ans. $a > 1$

Sol. -

Q10. Let a, b, c be real numbers, $a \neq 0$. If α is root of $a^2x^2 + bx + c = 0$. β is the root of $a^2x^2 - bx - c = 0$ & $0 < \alpha < \beta$, then show that the equation $a^2x^2 + 2bx + 2c = 0$ has a root γ that always satisfies $\alpha > \gamma > \beta$

Ans. -

Sol. -

Q11. If α, β are the roots of the equation, $(\tan^2 135^\circ)x^2 - (\operatorname{cosec} 10^\circ - \sqrt{3} \sec 10^\circ)x + \tan^2 240^\circ = 0$ then find the quadratic equation whose roots are $(2\alpha + \beta)$ and $(\alpha + 2\beta)$.

Ans. -

Sol. -

Q12. Let $x = \frac{1}{3}$ or $x = -15$ satisfies the equation, $\log_8(kx^2 + wx + f) = 2$. If k, w and f are relatively prime positive integers then find the value of $k + w + f$.

Ans. 96

Sol. -

Q13. If roots of the equation $x^2 - 10cx - 11d = 0$ are a, b and those of $x^2 - 10ax - 11b = 0$ are c, d, then find the value of $a + b + c + d$. (a, b, c and d are distinct numbers)

Ans. 1210

Sol. -

Paragraph

Consider the cubic equation

$$x^3 - (1 + \cos \theta + \sin \theta)x^2 + (\cos \theta \sin \theta + \cos \theta + \sin \theta)x - \sin \theta \cdot \cos \theta = 0$$

whose roots are x_1, x_2 and x_3 .

Q14. The value of $x_1^2 + x_2^2 + x_3^2$ equals

- (A) $2 \cos \theta$ (B) $\sin \theta(\sin \theta + \cos \theta)$ (C) 1 (D) 2

Ans. D

Sol. -

Paragraph

Consider the cubic equation

$$x^3 - (1 + \cos \theta + \sin \theta)x^2 + (\cos \theta \sin \theta + \cos \theta + \sin \theta)x - \sin \theta \cdot \cos \theta = 0$$

whose roots are x_1, x_2 and x_3 .

Q15. Number of values of θ in $[0, 2\pi]$ for which at least two roots are equal

- (A) 6 (B) 5 (C) 4 (D) 6

Ans. B

Sol. -

Paragraph

Consider the cubic equation

$$x^3 - (1 + \cos \theta + \sin \theta)x^2 + (\cos \theta \sin \theta + \cos \theta + \sin \theta)x - \sin \theta \cdot \cos \theta = 0$$

whose roots are x_1, x_2 and x_3 .

Q16. Greatest possible difference between two of the roots if $\theta \in [0, 2\pi]$ is

- (A) 2 (B) 1 (C) $\sqrt{2}$ (D) $2\sqrt{2}$

Ans. A

Sol. -

Q17. **Column-I**

Column-II

(A) If $\alpha, \beta \in (0, \pi)$ and $\alpha \neq \beta$ satisfy the equation $\frac{1 - \cos 2\theta}{\sin \theta} = \frac{1}{2}$,

(P) 0

then the value of $\tan\left(\frac{\alpha}{2}\right) + \tan\left(\frac{\beta}{2}\right)$ is equal to

(Q) 8

(B) If the expression $\frac{x^2 + (2m + 3)x + (m^2 + 3)}{\sqrt{x^2 + (2m + 1)x + m^2 + 2}}$

(R) 1

is non-negative $\forall x \in \mathbb{R}$, then the possible values of m can be equal to

(S) -1

(C) If the parabola $y = 5x^2 + x - 3$ lies above the

(T) 2

parabola $y = 2x^2 + 6x - 1$, then integral values of x can be equal to

(D) The number of real solutions of the equation $x^{2\log_x(x+3)} = 16$ is equal to

(A) (A) - Q (B) - PS (C) - QS (D) - P (B) (A) - Q (B) - P (C) - Q (D) - P

(C) (A) - Q (B) - PS (C) - S (D) - S (D) (A) - Q (B) - PS (C) - P (D) - QS

Ans. 5ee0675a86ce94b4ec0565f4

Sol. -

Q18.

Column-I

Column-II

(A) The expression, $x^{\ln y - \ln z} \cdot y^{\ln z - \ln x} \cdot z^{\ln x - \ln y}$

(P) $\frac{1}{8}$

when simplified reduces to

(B) Let $p = \log_5 \log_5(3)$. If $3^{C+5-p} = 5$, the value of C equals

(Q) $\frac{1}{4}$

(C) The real value of x for which the statement

(R) $\frac{1}{2}$

$\log_6 9 - \log_9 27 + \log_8 x = \log_{64} x - \log_6 4$ holds true, is

(S) 1

(D) $\frac{1}{2} \log \sqrt{5} \sin\left(\frac{\pi}{5}\right) \cdot \log_{\sqrt{\sin\frac{\pi}{5}}} 5$ simplifies to

(T) 2

(A) (A)- S (B)- P (C)- S (D)- T (B) (A)- S (B)- T (C)- P (D)- T (C) (A)- S (B)- P (C)- T (D)- S

(D) (A)- S (B)- P (C)- S (D)- T

Ans. 5ee070ef86ce94b4ec0566ce

Sol. -

Q19.

Column-I

Column-II

(P) If equation $x^2 + px + q = 0$ ($p, q \in \mathbb{R}$) and $x^3 + 3x^2 + 5x + 3 = 0$

(1) 1

has two common roots then $(p + q)$

(2) 2

(Q) If $f(x) = 4x^2 + 3x - 7$ and is common roots of equation $x^2 - 3x + 2 = 0$ and $x^2 + 2x - 3 = 0$ then value of $f(\alpha) =$

(3) 0

(R) Number of real roots of equation $\log_3 (x^2 + 9) + \log_3 (x^4 + 81) = 6$

(4) 5

(S) Number of real roots of equation $\log x(5x - 2) = 3$

(A) P Q R S
3 4 2 1

(B) P Q R S
4 3 1 2

(C) P Q R S
2 1 4 3

(D) P Q R S
1 2 4 3

Ans. 5ee075c42ce96db4ce8c8817

Sol. -